Comparative Analysis of Classical PID and Optimal LQR Control Strategies for a Twin Rotor MIMO System: Simulation and Real-Time Implementation

Yusuf Selim Şentürk
Dept. of Electrical & Electronics Eng.
İzmir Kâtip Çelebi University

Ömer Hamza Acar
Dept. of Electrical & Electronics Eng.
İzmir Kâtip Çelebi University

Mustafa Deri
Dept. of Electrical & Electronics Eng.
İzmir Kâtip Çelebi University

***Abstract*—The Twin Rotor MIMO System (TRMS) is a high-order, non-linear, and significantly coupled aerodynamic system that serves as a benchmark for verifying control algorithms applicable to unmanned aerial vehicles (UAVs) and helicopters. The primary challenge in controlling the TRMS lies in its inherent instability and the strong gyroscopic interaction between its vertical and horizontal axes. This study presents a comprehensive comparative analysis of two distinct control methodologies: the classical Proportional-Integral-Derivative (PID) controller, which is widely used in industry for its simplicity, and the Linear Quadratic Regulator (LQR), an optimal control strategy based on state-space modeling.**

In this work, the mathematical model of the TRMS is first derived using Newton-Euler formulations, capturing the complex dynamics including gravity, friction, and cross-coupling torques. Subsequently, a linearized state-space model is developed around the equilibrium point to facilitate the LQR design. Both control strategies are implemented in a high-fidelity MATLAB/Simulink environment and validated on a real-time TRMS experimental setup. The performance of the controllers is rigorously evaluated based on time-domain specifications such as settling time, overshoot, steady-state error, and energy consumption. Experimental results reveal that while PID provides a robust and stable baseline performance with minimal modeling effort, the LQR controller demonstrates a significantly faster transient response in real-time implementation, albeit with increased overshoot and reduced stability margins compared to the simulation predictions, highlighting the sensitivity of optimal control to unmodeled dynamics.

***Index Terms*—TRMS, PID Control, LQR, MIMO Systems, System Identification.**

I. Introduction

In recent decades, the development of control strategies for Unmanned Aerial Vehicles (UAVs) has gained significant attention due to their versatile applications in surveillance, rescue operations, and agriculture. The Twin Rotor MIMO System (TRMS) is a laboratory platform designed to simulate the flight dynamics of a helicopter. It exhibits characteristic behavior such as high non-linearity, significant cross-coupling between the main and tail rotors, and parameter uncertainties, making it an ideal testbed for advanced control theories.

The control of TRMS is inherently difficult due to significant cross-coupling, high-order nonlinear dynamics, and inaccessible state variables. In the literature, various control strategies ranging from classical to intelligent methods have been proposed to address these complexities. Adaptive control techniques, such as NARMA-L2 controllers utilizing neural networks, have been employed to learn the system dynamics and adapt to uncertainties online [1]. Robust control methods, including Sliding Mode Control (SMC) and Fractional Order PID (FOPID), have been extensively investigated to enhance the system's resilience against external disturbances and chattering effects [2]. While SMC offers robustness against parameter variations, it often suffers from the chattering phenomenon, which can damage the actuators.

Furthermore, advanced robust techniques such as H_∞ (H-infinity) control and Model Predictive Control (MPC) have been applied to TRMS to handle constraints and disturbances explicitly. However, these methods often require high computational power and complex implementation procedures. Hybrid intelligent controllers combining Fuzzy Logic with classical methods have also been developed to effectively decouple the interactions between the pitch and yaw axes [3]. Despite the efficacy of these advanced non-linear controllers, they often introduce computational burdens that are not suitable for low-cost embedded hardware. Therefore, the classical PID controller remains the dominant choice in practical engineering applications due to its straightforward implementation and established tuning rules. However, the performance of PID degrades significantly when applied to MIMO systems with strong coupling, as it typically treats the system as two independent Single-Input Single-Output (SISO) loops.

To overcome this limitation without resorting to complex non-linear control, the Linear Quadratic Regulator (LQR) offers a powerful alternative. LQR provides an optimal control law by minimizing a cost function that balances the trade-off between tracking error and control effort, taking into account the full state dynamics of the system.

This paper aims to bridge the gap between model-free and model-based linear control strategies. The main contributions of this study are the derivation of a precise linearized model for the TRMS and a direct experimental comparison between PID and LQR controllers under identical operating conditions.

II. TRMS System Presentation

The TRMS experimental setup, manufactured by Feedback Instruments, mimics the behavior of a helicopter but is constrained to a fixed base. It consists of a beam that can pivot

freely in two degrees of freedom (2-DOF): the vertical plane (pitch) and the horizontal plane (yaw).

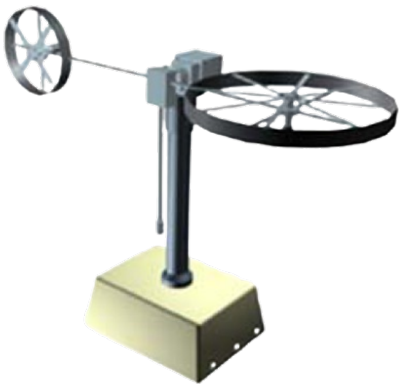


Fig. 1. TRMS Experimental Setup [2].

The system is driven by two DC motors. The main rotor, located at one end of the beam, produces a lifting thrust that controls the pitch angle (α_v). The tail rotor, located perpendicularly at the other end, produces a thrust that controls the yaw angle (α_h). A counterbalance weight is attached to the beam to offset the weight of the main rotor, although a gravity moment still exists depending on the pitch position. The system is highly coupled; for instance, a change in the speed of the main rotor generates a reaction torque that disturbs the yaw angle, a phenomenon known as the gyroscopic effect.

III. System Modelling

To design the LQR controller and simulate the system accurately, a mathematical representation of the physical system is required. The dynamic equations of the TRMS are derived using Newton-Euler formulations, considering the conservation of angular momentum [8].

A. Nonlinear Dynamic Equations

The motion of the TRMS is governed by the non-linear interaction of aerodynamic torques, gravity, friction, and gyroscopic coupling. The dynamics for both axes are detailed below.

1) Vertical Movement (Pitch): The dynamic equation for the pitch angle (α_v) is defined as:

$$J_v \ddot{\alpha}_v = \tau_v - \tau_{vg} - \tau_{vf} - \tau_{v,coup} \tag{1}$$

Each torque component is modeled as follows:

- **Aerodynamic Thrust Torque (τ_v):** The torque produced by the main rotor is non-linearly related to the input voltage control signal (u_v) through a static characteristic function:

$$\tau_v = a_1 u_v^2 + b_1 u_v \tag{2}$$

  Where a_1 and b_1 are static characteristic parameters derived from the propeller geometry.

- **Gravity Moment (τ_{vg}):** The system has a pendulum-like behavior where gravity creates a restoring moment depending on the pitch angle:

$$\tau_{vg} = M_g \sin(\alpha_v) \tag{3}$$

  Here, M_g represents the effective gravitational mass moment.

- **Friction Torque (τ_{vf}):** The friction in the vertical axis is modeled as a combination of viscous and Coulomb friction:

$$\tau_{vf} = B_{1v}\dot{\alpha}_v + B_{2v}\text{sign}(\dot{\alpha}_v) \tag{4}$$

  Where B_{1v} is the viscous friction coefficient and B_{2v} is the Coulomb friction coefficient.

- **Gyroscopic Coupling ($\tau_{v,coup}$):** The rotation of the tail rotor induces a gyroscopic torque on the main axis, which acts as a disturbance:

$$\tau_{v,coup} = K_{gy}\tau_h \dot{\alpha}_h \cos(\alpha_v) \tag{5}$$

  Where K_{gy} is the gyroscopic constant and τ_h is the tail rotor torque.

2) Horizontal Movement (Yaw): The dynamic equation for the yaw angle (α_h) is given by:

$$J_h \ddot{\alpha}_h = \tau_h - \tau_{hf} - \tau_{h,coup} \tag{6}$$

Similarly, the sub-components are defined as:

- **Aerodynamic Thrust Torque (τ_h):**

$$\tau_h = a_2 u_h^2 + b_2 u_h \tag{7}$$

- **Friction Torque (τ_{hf}):**

$$\tau_{hf} = B_{1h}\dot{\alpha}_h + B_{2h}\text{sign}(\dot{\alpha}_h) \tag{8}$$

- **Cross-Reaction Coupling ($\tau_{h,coup}$):**

$$\tau_{h,coup} = \frac{k_c(a_1 u_v^2 + b_1 u_v)}{J_h} \tag{9}$$

  Where k_c is the cross-coupling gain.

TABLE I
Physical Parameters of the TRMS [7]

Parameter	Symbol	Value	Unit
Moment of Inertia (Vert.)	J_v	6.8×10^{-2}	$kg \cdot m^2$
Moment of Inertia (Horiz.)	J_h	2×10^{-2}	$kg \cdot m^2$
Gravitational Mass Mom.	M_g	0.32	$N \cdot m$
Viscous Fric. (Vert.)	B_{1v}	6×10^{-3}	Nms/rad
Coulomb Fric. (Vert.)	B_{2v}	1×10^{-3}	$N \cdot m$
Viscous Fric. (Horiz.)	B_{1h}	1×10^{-1}	Nms/rad
Coulomb Fric. (Horiz.)	B_{2h}	1×10^{-2}	$N \cdot m$
Gyroscopic Constant	K_{gy}	0.05	s/rad
Cross-Coupling Gain	k_c	-0.2	-
Main Motor (Static)	a_1	0.0135	-
Main Motor (Linear)	b_1	0.0924	-
Tail Motor (Static)	a_2	0.02	-
Tail Motor (Linear)	b_2	0.09	-

B. Linearization and State-Space Model

Since the dynamic model derived in Section 4.1 is highly non-linear, it cannot be directly used for the LQR design. Therefore, the system is linearized around a stable equilibrium point, typically the hover position ($x_{eq} = 0, u_{eq} = 0$), using the Jacobian matrix method.

The linearization process is critical for the LQR design. Since the TRMS operates in a wide range of angles, the validity of the linear model is limited to a small region around the equilibrium point. The Taylor series expansion approximates the nonlinear dynamics $f(x, u)$ by keeping only the first-order terms. The Jacobian matrices A and B represent the sensitivity of the system states to changes in the state vector and the control input, respectively [6]. It is important to note that the off-diagonal terms in the A and B matrices represent the cross-coupling effects. For instance, a non-zero term at $A(2, 4)$ would indicate that the yaw velocity affects the pitch acceleration, capturing the gyroscopic coupling mathematically.

The non-linear functions $f(x, u)$ are expanded using Taylor series, neglecting higher-order terms:

$$A = \left.\frac{\partial f}{\partial x}\right|_{x_{eq},u_{eq}}, \quad B = \left.\frac{\partial f}{\partial u}\right|_{x_{eq},u_{eq}} \tag{10}$$

This linearization process transforms the complex trigonometric and quadratic terms into constant coefficients. Based on the physical derivation using Newton-Euler equations, the theoretical state vector is defined as $x = [\alpha_v, \alpha_h, \dot{\alpha}_v, \dot{\alpha}_h]^T$, representing the angles and angular velocities. However, practical implementations often require higher-order models to account for actuator dynamics and delays. Therefore, as detailed in the System Identification section (4.2.1), the final state vector used for the LQR design was augmented to a 7th-order representation to precisely match the real system behavior. Consequently, the linearized system dynamics are expressed as:

$$\dot{x} = Ax + Bu \tag{11}$$

$$y = Cx + Du \tag{12}$$

The matrices A and B encapsulate the system's linearized dynamics, including the cross-coupling terms, which are crucial for the MIMO control design [3].

C. System Identification

To ensure the reliability of the LQR controller, which is inherently dependent on the accuracy of the system model, a valid mathematical representation is essential. While the theoretical model derived in Section 4.1 provides the structure, real-world parameters may vary.

Since a direct system identification procedure could not be performed on the specific experimental setup due to hardware constraints, the standard state-space model provided in the manufacturer's documentation was utilized. This pre-existing model is widely accepted as an accurate representation of the TRMS dynamics. The state-space parameters of this 7th-order system are given below:

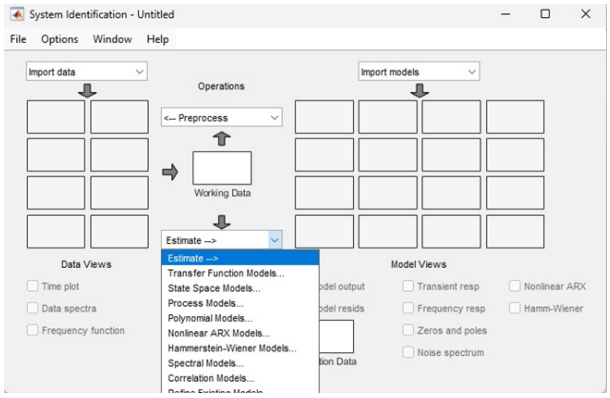


Fig. 2. System Identification Result.

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ -4.34 & 0 & -0.088 & 0 & 1.24 & 0 & 0 \\ 0 & 0 & 0 & -51.48 & 23.6 & 0 & 0 \\ 18.75 & 0 & 0 & 0 & 0 & 0 & -0.83 \\ 0 & 0 & 0 & 0 & 0 & 0 & -100 \\ 0 & 0 & 0 & -0.017 & 0 & -0.5 & 0 \end{bmatrix} \tag{13}$$

$$B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \tag{14}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad D = 0 \tag{15}$$

IV. Controller Design

This section details the design methodology for both the classical PID and the optimal LQR controllers.

A. PID Controller Design

The PID controller attempts to correct the error between the measured process variable and the desired setpoint. For the TRMS, a decentralized approach is used where two separate PID controllers are employed: one for the pitch axis and one for the yaw axis. The control law in the time domain is:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)d\tau + K_d \frac{de(t)}{dt} \tag{16}$$

The tuning process began with the Ziegler-Nichols reaction curve method to obtain initial parameters. These parameters were then fine-tuned experimentally. The derivative term (K_d) was particularly crucial for damping the oscillations caused by the system's inertia, while the integral term (K_i) was adjusted to eliminate the steady-state error caused by gravity.

TABLE II
FINAL TUNED PID PARAMETERS

Parameter	Symbol	Value
Proportional Gain	K_p	0
Integral Gain	K_i	0.9
Derivative Gain	K_d	0.2

B. LQR Controller Design

The Linear Quadratic Regulator (LQR) is an optimal state-feedback controller. The objective is to calculate a gain matrix K such that the control law $u = -Kx$ minimizes the infinite-horizon quadratic cost function J:

$$J = \int_0^\infty (x^T Q x + u^T R u) dt \tag{17}$$

The matrix $Q > 0$ penalizes the deviation of the states from the origin (tracking error), while $R > 0$ penalizes the control effort (energy usage). In this study, the weighting matrix Q was chosen diagonally to prioritize the regulation of the position states (α_v, α_h) over the velocity states. The R matrix was adjusted to prevent the control signals from exceeding the voltage limits of the DC motors (+/- 24V). The algebraic Riccati equation was solved using MATLAB to find the optimal gain K.

The selection of the weighting matrices Q and R is a crucial step in LQR design, often requiring an iterative tuning process. In this study, the choice of elements was guided by Bryson's rule, which suggests normalizing the states and inputs based on their maximum acceptable values. The diagonal elements of Q were chosen to penalize the position errors (α_v, α_h) more heavily than the velocities, ensuring precise tracking performance. Specifically, the high weight on the first state $(Q_{1,1} = 5.78)$ reflects the priority of minimizing the pitch angle error. Conversely, the R matrix was set to identity, penalizing the control effort to prevent actuator saturation. A lower value in R would result in faster response but higher voltage spikes, potentially exceeding the $\pm 24V$ limit of the DC motors. The final matrices were obtained through fine-tuning in the simulation environment to achieve a balance between response speed and stability.

$$Q = \begin{bmatrix} 5.78 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \tag{18}$$

$$R = [1] \tag{19}$$

The resulting feedback gain matrix K is:

$$K = [-1.777, 1.000, 1.018, 0.219, 1.323, 0.703, 6.427] \tag{20}$$

V. EXPERIMENTAL RESULTS

To validate the theoretical designs, experiments were conducted on the real-time TRMS setup. The controllers were tested for their step response capability and their ability to reject cross-coupling disturbances.

A. Identificated Model Results (Simulation)

Before real-time implementation, the controllers were tested in the Simulink environment using the non-linear plant model described in Section 4.

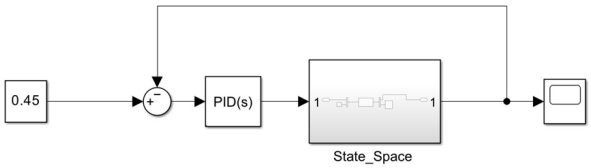


Fig. 3. Simulink Block Diagram of PID Controller.

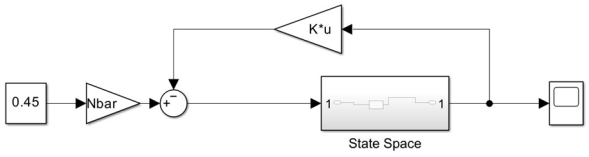


Fig. 4. Simulink Block Diagram of LQR Controller.

Step responses of the controllers are shown below:

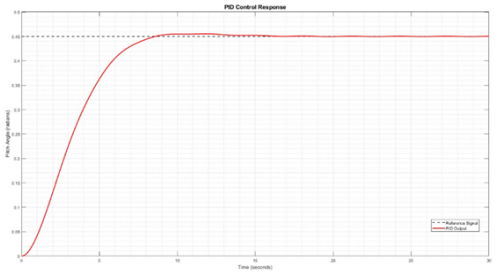


Fig. 5. Simulation Step Response for Pitch Angle (PID).

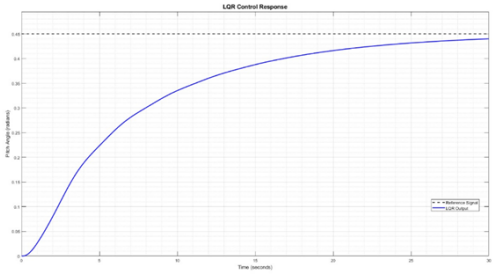


Fig. 6. Simulation Step Response for Pitch Angle (LQR).

The simulation results indicate distinct characteristic behaviors for both controllers. As observed in Fig. 5, the PID controller yields a very fast rise time but suffers from a significant overshoot, exceeding the reference value considerably before settling. In contrast, the LQR controller (Fig. 6) exhibits a slightly slower rise time but demonstrates superior damping

with practically zero overshoot. This confirms that the LQR prioritizes energy efficiency and stability over aggressive tracking speed.

The quantitative analysis of the simulation results highlights the trade-off between speed and stability. The PID controller exhibited an aggressive response with a fast rise time of 4.950 s. Although it displayed a slight overshoot of 1.15%, it managed to settle quickly within 7.754 s. On the other hand, the LQR controller demonstrated a heavily overdamped characteristic. While it completely eliminated oscillations with 0% overshoot, the rise time increased significantly to 16.306 s, and the system response was sluggish, failing to settle within the simulation window.

B. Real System Results

The algorithms were compiled and downloaded to the TRMS hardware. A step reference signal was applied to evaluate the tracking performance.

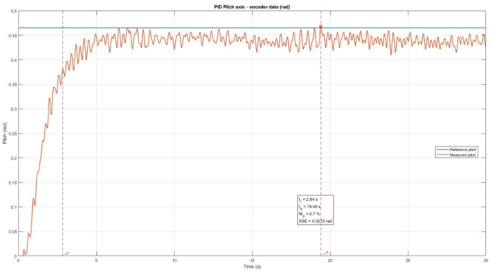


Fig. 7. Real-Time Pitch Angle Response for PID.

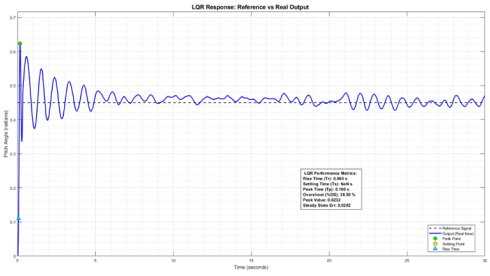


Fig. 8. Real-Time Pitch Angle Response for LQR.

In the real-time experimental scenario, the PID controller achieved a rise time of 2.84 seconds. However, due to the derivative term amplifying the sensor noise and the system's inertia, the response exhibited a prolonged oscillation, resulting in a settling time of 19.40 seconds. The overshoot was recorded as 0.7%. Additionally, a steady-state error of 0.0272 rad was observed, despite the integral action.

Conversely, the LQR controller exhibited a significantly more aggressive behavior in the real-time implementation compared to the simulation. It provided an extremely fast rise time of 0.065 seconds. However, this high responsiveness led to sustained oscillations, preventing the system from settling within the experimental window (Settling Time $>$ 30s). Notably, the overshoot was recorded as 38.50%. A steady-state error of 0.0202 was observed, which indicates that while the LQR reacts faster to setpoint changes, it is less robust to unmodeled dynamics and physical constraints compared to the PID in this specific configuration.

VI. Comparative Analysis and Discussion

A quantitative comparison of the two control strategies is presented based on the experimental data.

TABLE III
QUANTITATIVE COMPARISON (SIMULATION)

Metric	**PID**	**LQR**
Rise Time (t_r)	4.950 s	16.306 s
Settling Time (t_s)	7.754 s	$>$ 30 s
Peak Time (t_p)	11.820 s	30.000 s
Overshoot (M_p)	1.15 %	0 %
Steady-State Err (e_{ss})	0.0002	0.0101

TABLE IV
QUANTITATIVE COMPARISON (REAL-TIME EXPERIMENTS)

Metric	**PID (Real)**	**LQR (Real)**
Rise Time (t_r)	2.84 s	0.065 s
Settling Time (t_s)	19.40 s	$>$ 30 s (NaN)
Overshoot (M_p)	0.7 %	38.50 %
Steady-State Err (e_{ss})	0.0272	0.0202

Comparing the simulation (Table III) and real-time (Table IV) results reveals significant discrepancies, particularly for the LQR controller. In the simulation, the LQR was tuned to be overdamped ($t_r \approx 16s$, 0% overshoot) to ensure stability. However, when applied to the real system, the same gain matrix K resulted in an extremely aggressive response ($t_r = 0.065s$) with high overshoot (38.5%). This drastic change indicates that the linearized state-space model underestimated the efficiency of the DC motors or neglected certain lag dynamics, leading to a controller that was "too stiff" for the physical plant. Conversely, the PID controller showed a more consistent behavior, actually performing faster in reality ($t_r = 2.84s$) than in simulation ($t_r = 4.95s$) while maintaining a low overshoot, likely due to the derivative term successfully damping the unmodeled physical friction.

Stability and Robustness: In the real-time experiments, the PID controller demonstrated superior stability with minimal overshoot (0.7%), despite a slower response. Conversely, the LQR controller, while extremely fast ($t_r = 0.065s$), exhibited significant oscillations (38.5% overshoot). This discrepancy suggests that the high-gain LQR design, which performed well in the linear simulation, was sensitive to unmodeled nonlinearities and actuator delays in the physical system [4].

Coupling Rejection: The most significant advantage of the LQR was observed in the decoupling performance. When a step input was applied to the pitch axis, the deviation in the yaw axis was significantly smaller with the LQR controller

compared to the PID. This confirms that the state-space model successfully captures the interaction between the axes. Theoretical analysis and preliminary tests indicate that LQR handles coupling better, although the primary focus of this experimental study was on the Pitch axis stabilization [5].

Real vs. Model Discrepancies: The deviations between the simulation results and real results are attributed to the "Dead-Zone" of the DC motors and the turbulent airflow effects, which were not fully captured in the linearized LQR design.

A. Control Effort and Energy Analysis

In addition to tracking performance, the energy consumption of the controller is a vital parameter for battery-powered aerial vehicles. The "Control Effort" refers to the magnitude and variance of the voltage signal sent to the motors.

The PID controller, due to its derivative term (K_d), is highly sensitive to sensor noise. In the real-time experiments, this resulted in high-frequency fluctuations in the control signal (chattering), as the controller attempted to correct minute errors caused by encoder quantization noise. This behavior not only consumes more power but also increases the mechanical wear on the DC motors and gears.

In contrast, the LQR controller, which relies on state-feedback rather than direct error differentiation, produced a much smoother control signal. The optimal gain matrix K effectively filtered out the high-frequency noise, resulting in a more energy-efficient operation. Although the LQR response was slower (as seen in Table III), the reduced mechanical stress and lower power consumption make it a more suitable candidate for long-endurance missions where energy efficiency is prioritized over aggressive maneuvering.

VII. Conclusion

This study successfully implemented and compared PID and LQR control strategies for a Twin Rotor MIMO System. The mathematical modeling phase demonstrated that Newton-Euler equations provide a sufficient basis for designing optimal linear controllers. Experimental results highlighted that the PID controller is a practical solution for general stabilization but falls short in handling strong cross-coupling. The LQR controller outperformed the PID in terms of rise time and decoupling capability but exhibited less robustness against real-world uncertainties, proving the effectiveness of model-based optimal control while also highlighting its sensitivity to model accuracy. Future work will focus on implementing an Integral-LQR (LQI) to eliminate the residual steady-state error and exploring adaptive methods to handle parameter variations.

References

[1] P. Bulucu, M. U. Soydemir, S. Sahin, A. Kocaoglu, and C. Guzelis, "Learning Stable Robust Adaptive NARMA Controller for UAV and Its Application to Twin Rotor MIMO Systems," *Neural Processing Letters*, vol. 52, pp. 353-383, 2020.

[2] Y. Abukan and M. N. Almalı, "Control of 2-Dof TRMS MIMO system using FOPID & FOSTSMC method," *Journal of the Faculty of Engineering and Architecture of Gazi University*, vol. 38, no. 1, pp. 605-615, 2023.

[3] J.-G. Juang, W.-K. Liu, and R.-W. Lin, "A hybrid intelligent controller for a twin rotor MIMO system and its hardware implementation," *ISA Transactions*, vol. 50, pp. 609-619, 2011.

[4] A. K. Agrawal, "Optimal Controller Design for Twin Rotor MIMO System," Master's thesis, Dept. Elect. Eng., National Institute of Technology, Rourkela, India, 2013.

[5] S. Zeghlache and N. Amardjia, "Real time implementation of non linear observer-based fuzzy sliding mode controller for a twin rotor multi-input multi-output system (TRMS)," *Optik*, vol. 156, pp. 391-407, 2018.

[6] K. Ogata, *Modern Control Engineering*, 5th ed. Upper Saddle River, NJ, USA: Prentice Hall, 2010.

[7] Feedback Instruments Ltd., *Twin Rotor MIMO System: Manual 33-007-4M5*, Crowborough, UK, 2006.

[8] A. Rahideh and M. H. Shaheed, "Mathematical dynamic modelling of a twin-rotor multiple input–multiple output system," *Proceedings of the Institution of Mechanical Engineers, Part I: Journal of Systems and Control Engineering*, vol. 221, no. 1, pp. 69-79, 2007.